ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH, INDORE

DEPARTMENT OF INFORMATION TECHNOLOGY

AI मंथन 2.0

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Problem Statement 16: Multimodal Live Conversation Assistant for People with Speech Impairments

Domain: AI for Biomedical Engineering and Assistive Technology

Problem Statement: People with speech impairments caused by ALS, cerebral palsy, Parkinson's disease, stroke, or laryngectomy struggle to take part in live conversations: their speech is hard for standard systems to recognise, and voice-only tools ignore the gestures, facial expressions and lip movements they naturally use to communicate. Participants must build a real-time system that analyses the patient's voice together with gestures, facial expressions and lip movements from a camera, fuses them to infer the intended message, listens to the other person in the conversation, and helps the patient respond by suggesting replies and speaking the confirmed message in a natural, personalised voice. The use of ElevenLabs for speech recognition and voice generation is mandatory. The system should support English, Hindi and code-mixed speech, and enforce consent for any voice cloning.

Your Challenge: Design, train, and validate a robust prototype that successfully demonstrates end-to-end functionality, scalability, and real-world applicability within the hackathon timeframe.

Hints: Use ElevenLabs Speech-to-Text (Scribe) to transcribe both the patient and the other speaker, and measure where it fails on atypical speech. Extract hand, face and lip landmarks with tools such as MediaPipe, and fuse them with the transcript using an LLM that ranks candidate intents using conversation context. Use ElevenLabs voice cloning (or Voice Design when no recordings exist) with Flash or v3 models for fast, expressive replies. Start with TORGO or UA-Speech, or simulate atypical speech and gestures and label the data clearly as simulated. Evaluate using word error rate with and without gesture fusion, intent accuracy, end-to-end latency (target under 2 seconds), and effort needed from the user to confirm a reply. Use only consented voices.